

# AI-Driven Delinquency Risk Assessment Report

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## 1. Summary of Predictive Insights

The exploratory analysis and predictive modeling process identified several important indicators associated with increased delinquency risk. Customers with repeated late payments, high credit utilization, and elevated debt-to-income ratios demonstrated significantly higher probabilities of missing future payments. Additional patterns suggested that customers with unstable income or multiple recent credit applications may also require closer monitoring.

The predictive modeling framework recommended logistic regression as the primary model due to its transparency, interpretability, and suitability for binary risk classification. Decision tree analysis was also considered to improve explainability and support business understanding of key risk drivers.

### Top 3 Risk Factors

- Repeated missed or late payments
- High credit utilization ratio
- High debt-to-income ratio

### High-Risk Customer Segments

- Customers with two or more missed payments
- Customers using a high percentage of available credit
- Customers with unstable income or rising outstanding balances

## 2. Recommendation Framework

### SMART Business Recommendation

Launch a 6-week proactive SMS and email outreach program targeting customers with high credit utilization and multiple missed payments, with the goal of reducing 30+ day delinquency rates by 10%.

## Business Rationale

This recommendation is practical, scalable, and aligned with Geldium’s objective of improving collections efficiency through early intervention. Proactive communication can encourage timely repayments, improve customer engagement, and reduce financial losses before accounts become severely delinquent.

# 3. Ethical and Responsible AI Considerations

AI-driven financial decisions must remain fair, transparent, and explainable. While predictive models can improve risk assessment efficiency, they should support human decision-making rather than replace it entirely.

## Fairness Risks and Mitigation Strategies

| Fairness Risk                                                                                | Mitigation Strategy                                                                              |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Historical bias in customer data may unfairly classify certain customer groups as high-risk. | Regular fairness audits and demographic performance reviews should be conducted to monitor bias. |
| Certain variables may unintentionally act as proxies for protected characteristics.          | Sensitive or potentially biased variables should be reviewed and excluded where necessary.       |

## Explainability Approach

Model predictions should be communicated in simple business language. For example, customers may be classified as higher risk due to repeated missed payments, high outstanding balances, or elevated credit utilization levels. Using interpretable models such as logistic regression and decision trees improves transparency and stakeholder trust.

## Responsible AI Practices

The proposed approach promotes responsible AI adoption through fairness monitoring, explainability, privacy protection, and human oversight. Recommendations generated from predictive models should always be reviewed within business and regulatory contexts before implementation.

# Conclusion

The predictive modeling framework identified key delinquency indicators that can help Geldium improve collections prioritization and proactive customer engagement. By combining explainable AI models with responsible decision-making practices, Geldium can reduce delinquency risk while maintaining fairness, transparency, and customer trust. Early intervention strategies supported by AI-driven insights can improve operational efficiency and strengthen long-term financial outcomes.